Medical Assistive Robot

Human-Following Smart Dustbin.

Project title: Human-Following Smart Dustbin

Course Name: Robotics Lab

Course Code: CSE 438

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- ## Project Objective:

To develop a smart medical assistive robot that autonomously follows nurses or healthcare workers and automatically opens its lid when they bring medical waste close to the bin, ensuring safe, hygienic, and hands-free waste disposal in hospital environments.

- ## Project Goals:

Automate medical waste collection in hospitals.

Reduce human effort and cross-contamination risks.

Provide fast, safe, and hygienic waste disposal.

Help healthcare workers and patients avoid touching biohazardous waste.

Show how a microcontroller-based robot can work in real medical settings.

Improve patient safety through autonomous robot assistance.

➢ Executive Summary:

This project develops a smart assistive robot that can follow a user and support hygienic waste disposal. The system reads distance and sensor values, controls robot movement, and opens or closes the dustbin lid automatically. The main goal is to reduce manual contact and make waste disposal safer and easier.

➢ My Contribution in This Project:

My contribution was mainly in the coding and algorithm development part. I worked on:

- Developing the robot movement logic
- Writing the pseudocode
- Preparing the flowchart
- Creating the main decision-making algorithm
- Explaining how the robot stops, moves forward, turns, and controls the lid
- Preparing a code snippet to show the key programming logic

➢ pseudocode:

```
START robot

INITIALISE all sensors, motors, and servo

CLOSE dustbin lid

WHILE robot is running

    READ human distance
    READ left and right IR sensors

    IF human is too far OR too close
        STOP robot
    ELSE
        MOVE forward

    IF left sensor detects
        TURN left
    ELSEIF right sensor detects
        TURN right

    READ dustbin distance

    IF hand or waste is near
        OPEN lid
    ELSEIF lid is open for 3 seconds
        CLOSE lid

    REPEAT process

END
```

Algorithm Development:

Step 1: Start the robot program.

Step 2: Initialise the required sensors, motors, and servo.

Step 3: Close the dustbin lid at the beginning.

Step 4: Continuously read the human-following distance and IR sensor values.

Step 5: If the human is too far or too close, stop the robot. Otherwise, move forward.

Step 6: If the left sensor detects, turn left. If the right sensor detects, turn right.

Step 7: Read the dustbin distance.

Step 8: If hand or waste is detected near the dustbin, open the lid.

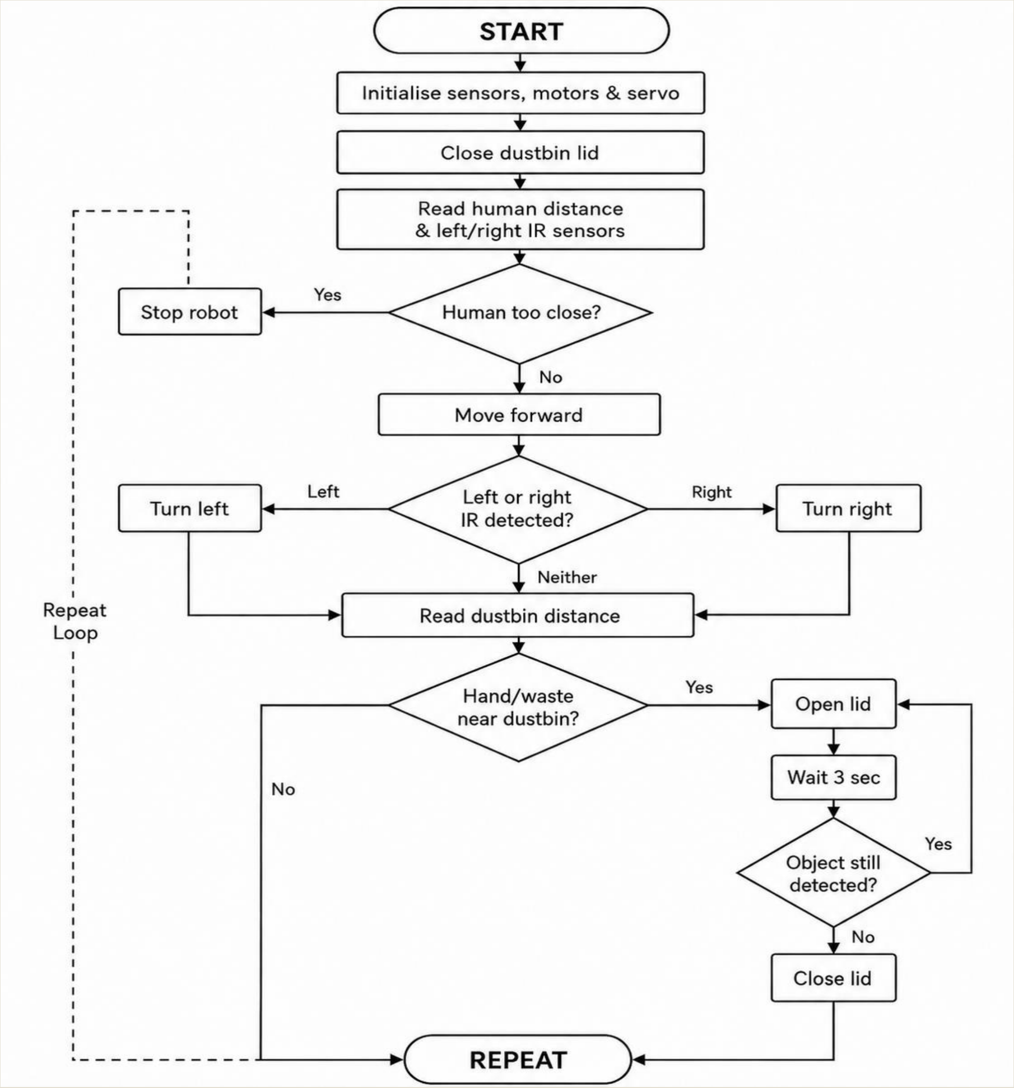
Step 9: If the lid remains open for 3 seconds and no object is detected, close the lid.

Step 10: Repeat the whole process continuously.

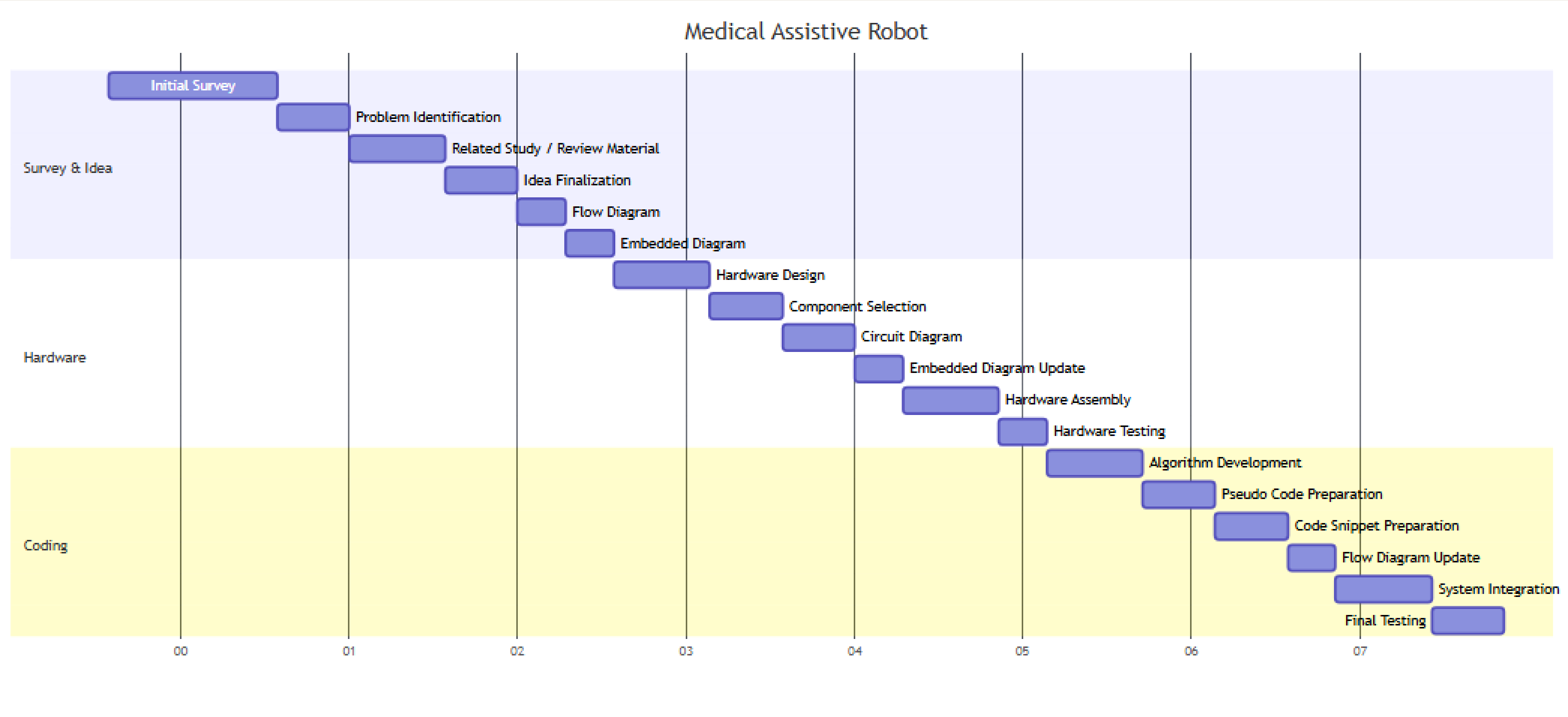
➢ Code Snippet:

```
void loop() {

  int distance = getFollowDistance();
  int leftIR = digitalRead(IR_LEFT);
  int rightIR = digitalRead(IR_RIGHT);

  if (distance > maxDist || distance <=
minDist) {
    stopCar();
  }
  else {
    forward();
  }

  if (leftIR == LOW && rightIR == HIGH) {
    turnLeft();
  }
  else if (leftIR == HIGH && rightIR ==
LOW) {
    turnRight();
  }

int dustbinDistance = getDustbinDistance();

  if (dustbinDistance <= 6) {
    dustbinServo.write(90);
    lidOpen = true;
    lastSeenTime = millis();
  }

  else if (lidOpen && millis() - lastSeenTime >= waitTime) {
    dustbinServo.write(0);
    lidOpen = false;
  }

  delay(100);
}
```

➢ Flowchart:


START
Initialise sensors, motors & servo
Close dustbin lid
Read human distance & left/right IR sensors
Human too close?
Yes
Stop robot
No
Move forward
Left or right IR detected?
Left
Turn left
Right
Turn right
Neither
Read dustbin distance
Hand/waste near dustbin?
Yes
Open lid
Wait 3 sec
Object still detected?
Yes
No
Close lid
No
Repeat Loop
REPEAT

➢ Grantt chart:



➢ Visual Proof:



Video link: https://drive.google.com/file/d/10ocNiq4UXC73CNx9Z_gX4YWCBoWxla1s/view?usp=sharing

Link 2: https://drive.google.com/file/d/1ktEx9mcCNPJyYcqycJBiPYat0uUjbrmE/view?usp=drive_link

- THANK YOU